Geodesic Projection: A Multi-Slot Compression Pipeline with Adaptive Phase-, Gauge-, Thermal-, and Drift-Aware Extensions

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Note

Project nomenclature. We use a four-level naming hierarchy throughout this paper series. *HyperTensor* is the project (compiler, runtime, kernels, theory). *Geodesic Projection (GP)* — the subject of this paper — is the inference-time pipeline that factorises every compressible weight along its dominant geometric directions. *Geodesic Runtime Compression (GRC)* is the attention-only subset of GP and is the empirical anchor of the companion paper. *OTT/GTC* is the Riemannian-manifold theoretical framework built on top of GP and is treated separately. In short: GRC $\subset$ GP $\subset$ HyperTensor, and OTT/GTC is the theoretical superstructure that explains why GP works.

Abstract

We describe *Geodesic Projection* (GP), the full multi-slot calibration-free compression pipeline implemented in the `geodessical` runtime, and four adaptive extensions that turn GP from a static rank-k scheme into a phase-, gauge-, thermal-, and drift-aware one. GP generalises the attention-only construction of the companion paper to all seven compressible weight matrices in a transformer block (Q, K, V, O, FFN up/gate/down) with a per-slot decomposition strategy chosen from the slot's measured singular spectrum, a manifold-curvature-driven per-layer rank allocation, and a persistent two-tier on-disk geometry + projected-weight cache validated by a single "depth-sink" layer hash. We then describe four orthogonal adaptive layers shipped on top of GP: (i) *MCR phase-aware rank allocation* that detects the empirical Mix/Compress/Refine layer phases and re-budgets rank; (ii) *Axiom Gauge*, an exact GL(d) residual-stream gauge optimisation that strictly lowers the total tail-energy of low-rank factorisations and is bakeable, zero-cost at inference; (iii) *Thermal Rank* and a tokens-per-joule diffplan term that couple rank to NVML-reported temperature and power; and (iv) *Online Basis*, a rejection-driven Oja [**oja1982**] update that revises the basis only when speculative-decode rejections expose drift. We present the cross-architecture activation-manifold evidence (intrinsic dim 11–25 on three open-weight models with $d \in \{576, 1536, 3072\}$) and full per-slot SVD spectra on Llama-3.1-8B that motivate the design, and we are explicit about which results are *measured* (the static-GP empirical anchor) vs. *design-validated* (the four adaptive extensions, whose first measurement sweep is the v0.2 milestone).

Keywords: transformer compression, low-rank factorisation, manifold geometry, residual-stream gauge, thermal-aware inference, online PCA, Oja's rule.

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Part I
The Geodesic Projection Pipeline

1 Introduction and Scope

The companion paper isolates a single design choice (calibration-free attention-only fixed-rank compression) and reports an end-to-end measurement on Llama-3.1-8B-Instruct Q4_K_M. Geodesic Projection is the *full* compression pipeline that the `geodessical` runtime implements. GP extends the companion-paper construction along three axes:

- **Slot coverage**: Q, K, V, O, FFN up/gate, and a dedicated SVD path for FFN down.
- **Per-layer rank**: driven by a manifold-curvature-ratio (MCR) heuristic with a hard floor.

- **Persistent caching**: a two-tier on-disk artifact (basis + projected weights) validated by the depth-sink layer's spectrum hash.

Two things this paper does *not* do, and the reader should treat them as scope limits. We do not provide multi-model end-to-end PPL/throughput sweeps—the empirical anchor remains the Llama-3.1-8B-Instruct Q4_K_M number set from the companion paper. We do not yet report Llama-3.1-70B numbers: compute is approved, the run is queued, and the pack is not in this revision. Throughout we use the term *design-validated* for code paths that exist, run without error, and are consistent with the implemented behaviour of their components, but for which no end-to-end benchmark has produced numbers.

2 Cross-Architecture Manifold Evidence

The premise of GP is that the local activation manifold of a trained transformer has intrinsic dimension much smaller than the ambient model dim d. If that premise is false, low-rank weight projection should be uniformly destructive. We ran the manifold-extraction pipeline of `legacy/axiom_vis/` (sampling → symmetry detection → Ricci curvature → axiom-set extraction) on three open-weight models.

Table 1: Intrinsic-dimensionality estimates (PCA, 95% variance retained) across three open-weight architectures. Source data: `legacy/axiom_vis/<model>/phase{1,4}_*.json`.

Model	d	k_{int}	k_{int}/d	Axiom set	Consistency
SmolLM2-135M	576	17	2.95%	24 / 96	0.921
Gemma-4-E2B	1,536	25	1.63%	23 / 92	0.961
Phi-3.5-mini	3,072	11	0.36%	22 / 96	0.959

The three values $\{11, 17, 25\}$ span an order of magnitude in d but do *not* grow with d. This is consistent with the isotropy/anisotropy literature on contextual embeddings and with the linear-representation hypothesis [**elhage2022superposition**]. Activation-space intrinsic dim is necessary but not sufficient for the GP construction; the actual weight-space evidence is per-slot SVD spectra (§ 3).

3 The Seven Slots and Their Spectra

The seven compressible matrices in a Llama-style block, with shapes for Llama-3.1-8B (d=4096, d_{ffn}=14336, GQA [**ainslie2023gqa**]): $Q, O : [4096, 4096]$; $K, V : [1024, 4096]$ (8 KV heads × 128); $W_{\text{up}}, W_{\text{gate}} : [14336, 4096]$; $W_{\text{down}} : [4096, 14336]$.

Table 2: Singular-spectra summary for all seven slots, layer-mean over $\{0, 7, 15, 23, 31\}$ on Llama-3.1-8B Q4_K_M (dequantised).

Slot	Shape	Rank for 95%	Rank for 99%	Energy@k=1024
Q	[4096, 4096]	1,682	2,434	0.836
K	[1024, 4096]	605	802	1.000 (GQA cap)
V	[1024, 4096]	809	954	1.000 (GQA cap)
O	[4096, 4096]	2,118	2,868	0.743
W_{gate}	[14336, 4096]	3,271	3,840	0.539
W_{up}	[14336, 4096]	3,360	3,873	0.492
W_{down}	[4096, 14336]	3,345	3,863	0.490

Asymmetry. Attention slots are highly compressible ($k_{95} \approx 0.4$–$0.5\,d$) and retain ~80% of energy at k=1024. FFN slots require $k_{95} \approx 0.8\,d$ and lose ~50% of Frobenius energy at k=1024. This is the

quantitative version of the qualitative "FFN is flat" claim [**geva2021ffn**]. We give W_{down} a dedicated SVD path rather than the Gram-eigendecomposition route used for the other six slots: the construction is mathematically equivalent in the rank-truncated limit but numerically more stable when the spectrum is approximately flat [**golub2013matrix**].

Geometric reading of the asymmetry. Attention and FFN solve different problems. The attention block is a routing circuit: $\mathbf{W}_Q, \mathbf{W}_K$ project tokens into a similarity space whose discriminative directions are concentrated in a few hundred coordinates (head subspaces plus the residual stream's preferred basis), and $\mathbf{W}_V, \mathbf{W}_O$ assemble a context-conditional message in that low-rank space. Routing is structurally low-rank because the number of "things being routed about" is small relative to d. The FFN, by contrast, behaves as a key-value memory bank [**geva2021ffn**]: each FFN column responds to a localised input pattern, so a high-quality FFN must span a high-dimensional set of keys. Removing FFN directions removes individual memories; removing attention directions removes (at worst) routing fidelity, partially recoverable from the residual stream. This predicts both the empirical $\sim 3.4\times$ rank gap between FFN and attention k_{95}/d and the runtime observation that aggressive FFN low-rank truncation is unstable at 8B scale while attention truncation is benign down to $k{=}1024$. GP therefore treats W_{up} and W_{gate} with a high-floor strategy and isolates W_{down} on its dedicated SVD path.

4 Per-Layer Rank Selection (MCR)

Constrained-budget formulation. Let layer $\ell \in \{1, \ldots, L\}$ have spectrum $\{\sigma_{\ell,1} \geq \sigma_{\ell,2} \geq \cdots\}$ and define the captured Frobenius energy at rank k_ℓ as

$$\mathcal{E}_\ell(k_\ell) \;=\; \frac{\sum_{i \leq k_\ell} \sigma_{\ell,i}^2}{\sum_i \sigma_{\ell,i}^2} \;\in [0, 1]. \tag{1}$$

GP allocates rank by approximately solving the budget-constrained maximisation

$$\max_{\{k_\ell\}} \sum_{\ell=1}^{L} \mathcal{E}_\ell(k_\ell) \quad \text{s.t.} \quad \sum_{\ell=1}^{L} k_\ell \leq K_{\text{budget}}, \;\; k_{\text{floor}} \leq k_\ell \leq K_{\max}. \tag{2}$$

Because each $\mathcal{E}_\ell$ is concave and non-decreasing, the Lagrangian KKT conditions reduce to a water-filling rule: spend the next rank slot on whichever layer has the largest marginal energy gain $\sigma_{\ell,k_\ell+1}^2 / \sum_i \sigma_{\ell,i}^2$. The MCR heuristic below is a closed-form proxy for this rule that uses the manifold curvature ratio in place of an expensive per-step spectrum lookup.

A naive scheme assigns the same k to every layer. GP exposes a manifold-curvature-ratio heuristic that allocates more rank to layers whose local activation geometry has higher (Ricci-style) curvature: $k_\ell = \text{clip}\big(\text{round}(k_{\text{target}} \cdot \text{MCR}_\ell / \overline{\text{MCR}}),\; k_{\text{floor}},\; K_{\max}\big)$ with $K_{\max}{=}1536$ (`AXEX_MANIFOLD_K_MAX`) and $k_{\text{floor}} \approx 0.55\,d$. The floor matters more than the cap; below $\approx 0.4\,d$ on small models we observe text-quality collapse.

Note

The MCR-driven allocation is the part of GP we are *least* confident about; it is enabled by default but the companion-paper measurements all use shared-rank for a clean calibration-free claim. An MCR vs. shared-rank A/B at matched mean-k is in the v0.2 milestone.

4.1 Sink-Basis Bypass

StreamingLLM [**xiao2023streamingllm**] preserves a handful of *sink tokens* in the KV cache because the first few positions hoard disproportionate attention mass and L2 norm. To our knowledge, no prior work preserves the corresponding *sink direction* in the weight/activation PCA basis. GP does.

Let $\hat{s} \in \mathbb{R}^d$ be the unit vector of the dominant residual-stream direction at the sink position (we estimate it as the top eigenvector of $\mathbb{E}[h_0 h_0^\top]$ over a calibration-free 32-token zero-prompt forward). Given the per-layer projector $V_r^{(\ell)}$ produced by MCR, define the *sink coverage*

$$\eta_\ell \;=\; \max_{1\le k\le r} \left|\langle \hat{s},\, V_r^{(\ell)}[:,k]\rangle\right|. \tag{3}$$

Whenever $\eta_\ell < \varepsilon_{\text{sink}}$ (default $\varepsilon_{\text{sink}} = 0.15$) the runtime appends $\hat{s}$ as an extra column and re-orthonormalises:

$$V_{r+1}^{(\ell)} \;\leftarrow\; \mathrm{QR}\big([\, V_r^{(\ell)} \;\; \hat{s}\,]\big). \tag{4}$$

This adds at most one rank per layer and bounds the sink-step reconstruction error by $\|h_0\| \sin\theta(\hat{s}, V_{r+1}^{(\ell)})$, which is zero by construction. Empirically the bypass fires on 3−7% of layers and eliminates the rare "first-token spike" in perplexity that shared-rank GP otherwise exhibits at very low k.

5 Geometry Cache and Depth-Sink Shortcut

Building GP bases from scratch on Llama-3.1-8B takes ~70 s on the reference GPU; on the 70B target it is extrapolated at ~70 min. The runtime persists two artifacts:

1. `ott_geometry.bin`: per-layer spectra, axiom set, depth-sink index, intrinsic-dim estimate, weight-blob hash; a few MB.
2. W_{proj} cache: projected weights, ~ 1,093 MB on Llama-3.1-8B at k=1536.

Both caches are keyed on $(\text{model digest}, \text{flag set}, k)$.

The Two-Thirds Dimensionality Saturation Rule. Reloading the full geometry cache on a 70B model is not free; we want only to verify the cache is not stale. The runtime defines the *depth-sink* as the single layer where the cumulative explained-variance curve of the residual stream first flattens to within 10^{-3} of its plateau. Across the four open transformers we have inspected this layer sits at depth $\ell^\star \approx 2L/3$:

model	layers L	depth-sink $\ell^\star$
Llama-3.1-8B	32	21
Gemma-2-2B	26	19
Phi-3.5-mini	32	22
Qwen-2.5-7B	32	22

We call this empirical observation the **Two-Thirds Dimensionality Saturation Rule** and exploit it as an $O(1)$ proxy for the otherwise $O(L)$ cache-integrity check: the integrity probe reads only the depth-sink layer's spectrum and recomputes it on the fly (< 200 ms on Llama-3.1-8B, vs. ~70 s for the full sweep). The rule is a property of these four models, not a theorem; we expect it to hold approximately on architectures whose residual stream is rank-saturating, and to break on mixture-of-experts or extremely shallow stacks.

6 Empirical Anchor: Static-GP Numbers

GP at attention-only-shared-rank *is* the companion-paper configuration, so its end-to-end numbers are reproduced here under the GP framing for self-containment. Hardware: RTX 4070 Laptop, 32 MB L2.

Table 3: End-to-end GP numbers on Llama-3.1-8B-Instruct Q4_K_M.

Configuration	Decode (%base)	PPL	W_{proj} disk
Baseline	100.00	6.7902	—
GP attn-only k=1024	106.27	10.96 (+61.4%)	729 MB
GP attn-only k=1536	97.55	7.69 (+13.30%)	1,093 MB
GP attn-only k=2048 requested	101.04	7.69 (+13.30%)	1,093 MB

The k=2048 row produces *identical* PPL to k=1536 because `AXEX_MANIFOLD_K_MAX` silently caps the stored basis; it also matches the disk footprint.

Part II
Adaptive Extensions

The four mechanisms below address four distinct invariance failures of static rank-k GP, are mutually orthogonal in their state, and each is zero- or near-zero-overhead at inference time. The contribution is the *combination*; each individual mechanism has antecedents in the literature.

Table 4: Four invariances assumed by static rank-k, their empirical violations, and the GP extension that addresses each.

Invariance	Violation	Mechanism
All layers compress equally well at the same rank.	Three measurable phases (Mix / Compress / Refine) with order-of-magnitude variance differences.	§ 7
The SVD spectrum is intrinsic to W.	The residual stream carries an exact $\mathrm{GL}(d)$ gauge; the spectrum of $W\,\mathrm{diag}(g)^{-1}$ depends on g.	§ 8
Optimal rank is hardware-independent.	Consumer GPUs throttle at $\sim$ 85 °C within $\sim$ 30 s; J/tok doubles in the throttled regime.	§ 9
Calibration distribution matches inference distribution.	Speculative-decode rejections directly measure drift.	§ 10

7 MCR Phase-Aware Rank Allocation

For each layer ℓ, compute the per-feature variance of captured hidden states $\sigma_\ell^2 = \mathbb{E}[\|h_\ell - \bar{h}_\ell\|^2]/d$, smooth with a 3-tap moving average $\tilde{\sigma}_\ell^2$, and let $\sigma_{\min}^2 = \min_\ell \tilde{\sigma}_\ell^2$. The phase label is

$$\mathrm{phase}(\ell) = \begin{cases} \text{Compress} & \tilde{\sigma}_\ell^2 \le c_{\text{thr}}\sigma_{\min}^2 \\ \text{Mix} & \ell < \ell_{\text{compress_start}} \\ \text{Refine} & \ell > \ell_{\text{compress_end}} \end{cases}$$

with default $c_{\text{thr}} = 1.5$. Given a global budget $R = r_{\text{base}} \cdot L$,

$$r_\ell = \text{clip}\big(s_{\text{phase}(\ell)} \cdot r_{\text{base}},\ r_{\min},\ r_{\max}\big),$$

default scales s_{Mix}=1.5, s_{Compress}=0.35, s_{Refine}=1.2, then renormalised so total budget matches R.

Attention-sink bypass. Sink tokens (BOS in particular) carry residual-stream norms several σ above the mean. StreamingLLM [**xiao2023streamingllm**] preserves sink tokens in the KV cache; we preserve the sink *direction* in the basis itself. When $\max_k \langle \hat{s}, v_k \rangle$ is small for the dominant sink direction $\hat{s}$, we append $\hat{s}$ to V_r before initialisation, costing one rank slot per affected layer.

Predicted gain. At fixed total budget the predicted PPL improvement scales with $\sigma^2_{\text{Mix}}/\sigma^2_{\text{Compress}}$, typically $\sim 5\times$ in published variance profiles. Whether the prediction holds is the v0.2 measurement.

8 Axiom Gauge: $\text{GL}(d)$ Gauge-Optimal Compression

The transformer residual stream carries an exact gauge symmetry. For any invertible $G \in \text{GL}(d)$, the substitution $x \mapsto Gx,\ W^{\text{read}} \mapsto W^{\text{read}}G^{-1},\ W^{\text{write}} \mapsto GW^{\text{write}}$ leaves outputs unchanged, but the SVD spectrum of $W^{\text{read}}G^{-1}$ depends on G unless G is orthogonal. Static rank-r schemes do not exploit this freedom.

Construction. Restrict to diagonal $G = \text{diag}(g)$, $g \in \mathbb{R}^d_{>0}$, parameterise $\lambda = \log g$ to keep $g > 0$, and minimise the joint tail energy

$$\mathcal{L}(g) = \sum_{\ell, W \in \text{reads}_\ell} \text{tail}_r\big(W\,\text{diag}(g^{-1})\big) + \sum_{\ell, W \in \text{writes}_\ell} \text{tail}_r\big(\text{diag}(g)\,W\big),$$

where $\text{tail}_r(M) = \|M\|_F^2 - \|\text{trunc}_r(M)\|_F^2$. The gradient in log-space is sparse and analytic:

$$\partial_{\lambda_i} \text{tail}_r(W\,\text{diag}(e^{-\lambda})) = -2\,\|X_{\text{tail}}[:, i]\|^2, \quad \partial_{\lambda_i} \text{tail}_r(\text{diag}(e^{\lambda})W) = +2\,\|X_{\text{tail}}[i, :]\|^2.$$

Both follow from $\|M\|_F^2 = \text{tr}(M^\top M)$. 10–30 outer iterations suffice; we normalise so $\prod_i g_i^{1/d} = 1$ to remove the global scale.

Bakeable, zero-overhead inference. Once $g^\star$ is found it is baked into the compressed factors before GPU upload: stored $\widetilde{V}_t[k][i] = S_k V_{k,i}\, g_i$ (read), $\widetilde{U}[i][k] = U_{i,k}/g_i$ (write). The inference path—the two-GEMV tmp $= V_r^\top x$, out $= U_r$tmp—is unchanged. This is the cleanest possible deployment: a free pre-compute step at calibration time, no runtime branch, no extra memory traffic.

Predicted gain. For models with LayerNorm-induced uniform feature scales the gain is modest ($\lesssim 5\%$ tail energy); for models with strong per-feature scale asymmetry, $\geq 20\%$ is plausible.

9 Thermal Rank and Tokens-per-Joule

On the reference RTX 4070 Laptop, sustained Llama-3.1-8B inference reaches $\sim 85\,^\circ$C within ~ 30 s and the boost clock is clamped; tok/s falls to 50–60% of cold-start. Earlier work clamps the clock proactively; we clamp the *compression rank* instead.

Linear interpolation rule. With temperature thresholds T_{low}=65 °C, T_{high}=85 °C and rank bounds $r_{\min}, r_{\max}$,

$$r(T) = \text{clip}\left(r_{\max} - (r_{\max} - r_{\min}) \cdot \frac{T - T_{\text{low}}}{T_{\text{high}} - T_{\text{low}}},\ r_{\min}, r_{\max}\right).$$

NVML is loaded dynamically; if unavailable the rank reverts to base. Polling is rate-limited to once per 500 ms. An optional power cap P_{budget} further reduces rank when current draw exceeds the cap. The system is closed-loop stable as long as the heat-generation rate at $r_{\min}$ is below the cooling capacity at T_{high}.

Tokens-per-joule diffplan term. The differentiable rank plan of [**robbins1951sa**] style minimises reconstruction error with an ℓ_1 rank penalty. We add an explicit energy term: with $J = P_{\text{NVML}}/(\text{tok/s})$ joules/token and a slowly-estimated coefficient $\hat{c}_{\text{rank}}$ (joules per unit-rank per token),

$$\Delta_{\text{TPJ}}\, \nabla_\ell^{(r)} = \lambda\, \hat{c}_{\text{rank}}\, p_\ell^{(r)} \big(R^{(r)} - r_\ell^{\text{soft}}\big), \quad r_\ell^{\text{soft}} = \sum_r p_\ell^{(r)} R^{(r)}.$$

The plan-level objective becomes $\mathcal{L}_{\text{recon}} + \alpha\, \|r\|_1 + \lambda\, \hat{c}_{\text{rank}}\, r^{\text{soft}}$—a weighted blend of accuracy, headroom, and energy.

Predicted gain. At r=1024 baseline tok/s drops $\sim$ 40% once throttling engages. Thermal Rank avoids throttle altogether at the cost of a controlled rank reduction; if rank–PPL elasticity is $< 1\%$ per rank-step in the operating regime (typical for $r \in [768, 1280]$ on Llama-3.1-8B), the trade is favourable.

10 Online Basis: Rejection-Driven Oja

A static PCA basis fitted on calibration data drifts out of validity as inference moves into uncovered distributions. Speculative decoding [**leviathan2023speculative**, **chen2023speculative**] provides a free measurement of drift: every *rejection* is a position where draft and verifier disagreed, so the residual $h^{\text{tgt}} - h^{\text{drft}}$ is a sample of the direction the current basis fails to capture.

Decayed Oja update. Write the projection basis at step t as $\mathbf{P}_t = W_t W_t^\top \in \mathbb{R}^{d\times d}$, the orthogonal projector onto the row-span of W_t. Oja's rule lifts this to a per-row update on W_t itself: for layer ℓ with basis $W_t \in \mathbb{R}^{k\times d}$ and sample $x = h^{\text{tgt}} - h^{\text{drft}}$,

$$w_i^{(t+1)} \leftarrow w_i^{(t)} + \eta_t\, x\, (x^\top w_i^{(t)}), \qquad w_i^{(t+1)} \leftarrow w_i^{(t+1)} \Big/ \Big\| w_i^{(t+1)} \Big\|, \quad i = 1, \ldots, k, \tag{5}$$

deflated row-wise so the matrix remains orthonormal. The induced map on the projector is $\mathbf{P}_{t+1} = \mathbf{P}_t + \eta_t \big(xx^\top - \mathbf{P}_t xx^\top \mathbf{P}_t\big) + O(\eta_t^2)$, i.e. a rank-one tilt of $\mathbf{P}_t$ *toward* the residual direction x at rate η_t. The learning-rate schedule $\eta_t = \eta_0/\sqrt{t}$ satisfies the Robbins–Monro conditions $\sum \eta_t = \infty$, $\sum \eta_t^2 < \infty$ [**robbins1951sa**]; default $\eta_0 = 0.01$. Convergence to the leading eigenvectors of the running covariance is classical [**oja1982**]; what is new is the *trigger*—only rejection residuals are fed in, by construction the directions the basis fails to capture.

Per-layer staleness. A bump-counter `onb_layer_state_t::version` is incremented on every applied update. GP tracks the version it last consumed; if a layer's version moves ahead, the cached projection is recomputed lazily on the next forward. Updates apply between decode steps, queue bounded to 256, with a 4-rejection minimum gate.

Predicted behaviour. On matched calibration/test, online updates should be a no-op. On linearly drifting OOD prompts, the Oja-decayed estimator converges to the running covariance with bias $\propto \eta_t$; for η_0=0.01, t=10^4 samples the bias floor is $\sim 10^{-4}$, well below typical PCA truncation error.

Part III
Composition, Limitations, Reproduction

11 How the Four Mechanisms Compose

The four extensions touch disjoint parts of the pipeline: MCR changes the per-layer r_ℓ *before* basis construction; Axiom Gauge changes the basis itself by left-/right-conjugating with $\mathrm{diag}(g)$; Thermal Rank changes the truncation threshold at *inference* time; and Online Basis changes the basis between decode steps in response to drift. Their state is non-overlapping (rank vector, gauge vector, thermal scalar, version counter), and their update intervals are very different (one-shot, one-shot, 500 ms, per-rejection). All four are runtime-flag-gated and individually disabling.

12 Limitations

- The static-GP empirical anchor is single-model, single-GPU, single quantisation (Llama-3.1-8B-Instruct Q4_K_M, RTX 4070 Laptop, dequantised to fp32). No cross-hardware or cross-model end-to-end measurement is included for the adaptive layers.
- All four adaptive extensions are *design-validated*: code paths exist, runtime flags work, but the v0.2 measurement sweep is the next milestone. Predicted gains in §§ 7 to 10 are upper-bound estimates from closed-form arithmetic, not measurements.
- The W_{proj} cache is fp32; converting to Q8_0 [**ggerganov2023kquants**] would close the disk-footprint gap but re-quantises an already-quantised weight, and behaviour at the second quantisation boundary is uncharacterised.
- Cache invalidation forces a full rebuild ($\sim$ 70 min for the 70B target), making model upgrades an event rather than a transparent rolling deploy.
- FFN-only configurations are observably slower than baseline at k=1024; the FFN spectrum is too flat for the attention-only construction to be a net win. We have not characterised this with a full sweep.

13 Reproduction

Static-GP empirical anchor: see the companion paper's reproduction recipe at https://github.com/NagusameCS/HyperTensor/blob/main/repro/REPRODUCE.md. Cross-model intrinsic-dim numbers are reproducible via:

```
# intrinsic-dim re-run for any model under legacy/axiom_vis/
geodessical <model.gguf> --axex-axiom-only --axex-export-vis <outdir>
# outputs <outdir>/phase1_manifold.json
```

The four adaptive extensions are reachable via: `-axex-mcr`, `-axex-gauge`, `-axex-thermal-rank`, `-axex-online-basis`, with sub-flags documented in `runtime/nn/axiom_exploit.h`.

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